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DIGITAL FARMERS' PRODUCE SYSTEM: A CLOUD-BASED PLATFORM FOR OPTIMIZING CROP MANAGEMENT AND MARKET LINKAGES

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ABSTRACT

Agriculture remains a primary source of livelihood in the Municipality of Sinacaban, Misamis Occidental, where many farmers rely on seasonal crops. Interviews with local farmers and the Municipal Agriculture Office revealed recurring problems such as lack of timely price information, weak linkages with buyers, delays in transactions, and limited visibility of available produce at the barangay level. These challenges contribute to low profitability and underutilized harvests.

This project develops the Digital Farmers' Produce System, a cloud-based web application designed to help farmers manage their seasonal crop listings, request support from the local agriculture office, and connect with potential buyers. The system allows farmers to publish their available produce with quantity and harvest schedules, buyers to search and view listings by location and category, and agriculture officers to monitor barangay-level stock information and resource requests.

The system is being developed using the Rapid Application Development (RAD) framework. User requirements were gathered through interviews and surveys with farmers, buyers, and agriculture staff. Wireframes and mock-ups were created to refine the user interface before implementing the system using PHP 8, the Flight microframework, MySQL, HTML, CSS, JavaScript, and Bootstrap on a shared cloud hosting environment. Security measures such as role-based access, encrypted communication, and validation of inputs are incorporated into the design.

A comprehensive plan for user testing and evaluation has been prepared. Functional, usability, and security testing will be conducted on a staging server, alongside a user acceptance survey based on the Technology Acceptance Model (TAM). The Digital Farmers' Produce System is expected to demonstrate how a localized, user-friendly, cloud-based platform can support municipal-level agricultural management and strengthen connections between farmers, buyers, and the local government unit.

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INTRODUCTION

Background of the project

Agriculture continues to be the main livelihood in the Municipality of Sinacaban, Misamis Occidental, where many farmers rely on seasonal crops as their primary source of income. Based on interviews with two active farmers in the area, they face recurring issues that limit productivity and profitability. These include difficulties in finding consistent buyers, a lack of timely price information, and losses caused by delayed transactions and poor market coordination (Dominguez, 2025).

These local challenges reflect national agricultural issues. (Mopera, 2016) reported that post-harvest losses in the Philippines can reach up to 42% for vegetables and 28% for fruits, largely because of inadequate logistics and limited market linkages. Likewise, (Briones, Galang, & Latigar, 2023) found that smallholder farmers continue to struggle with information asymmetry and weak digital connectivity, preventing them from maximizing their earnings and planning effectively for the next planting season.

The problem is important because it directly affects farmers' livelihoods and the stability of the food supply chain. Addressing it through digital innovation can strengthen rural economies, reduce waste, and encourage young people to engage in agriculture.

Several initiatives have been developed to address similar concerns. Programs like e-Kadiwa and the Department of Agriculture's eMarketplace were designed to connect farmers with consumers online. However, these national platforms remain underutilized in rural communities due to connectivity limitations, low digital literacy, and lack of localized features (Briones et al., 2023; Pastolero & Sassi, 2022). To solve these issues, the proposed web-based platform will use simple and easy to understand features. It will also include localized crop information and offer basic user training to help farmers use the system effectively.

The proposed project, *Sinacaban Digital Farmers' Seasonal Produce System*, addresses this gap by developing a cloud-based platform that helps farmers manage seasonal crops efficiently while strengthening their connection with buyers. The system will allow users to record harvest schedules, monitor seasonal trends, and access real-time buyer offers and market prices within one unified platform. Using the Rapid Application Development (RAD) methodology, the project will iteratively design, prototype, and test the system with actual users to ensure functionality and adoption. The expected outcome is an accessible, scalable system that reduces post-harvest losses, increases farmer income, and enhances local agricultural efficiency.

Objectives

The general objective of the project is to develop a web application that facilitates transactions between farmers and local government units' agriculture offices, as well as between farmers and buyers of seasonal crop produce. Specifically, the project aims to:

- Develop a feature where farmers can post their available produce.
- Enable the farmers to request necessary farming resources from the LGU's agriculture office through the web application.
- To provide buyers with an accessible platform to view available crops in their area and contact farmers directly.
- To enable the Sinacaban Department of Agriculture in Sinacaban to monitor crop stocks, respond to inquiries, and manage subsidies through a web-based application that supports informed policy and distribution decisions.
- To provide basic summary views and simple indicators that can support decision-making on crop trends and stock status.
- To develop a user-centered web application system using the Rapid Application Development (RAD) model, ensuring iterative feedback and rapid prototyping for effective user adoption.
- To evaluate the system's functionality, usability, and performance in enhancing

market linkages, improving productivity, and reducing post-harvest losses using a Likert-scale questionnaire assessed for reliability through Cronbach's alpha.

Scope and Limitations

Scope

The project aims to develop a web-based platform specifically designed for farmers in Sinacaban. Initially, the system will target smallholder farmers registered with the Municipal Agriculture Office and will focus on seasonal crops commonly grown in the municipality. Its main features include product posting, resource requests, buyer communication, and monitoring tools for the Department of Agriculture. The system will include role-based access: farmers, buyers, and agricultural officers. It will run on a cloud infrastructure to allow online access anytime and anywhere. Development will prioritize responsive design, basic analytics for stock and transaction tracking, and secure user authentication.

Limitations

The system will not include online payment or delivery features, as transactions between farmers and buyers will still occur offline. Internet connectivity in rural areas may affect access and performance. The accuracy of data will depend on the information entered by users. The system's initial scope will cover Sinacaban only, though it may be expanded to nearby municipalities in future versions.

REVIEW OF EXISTING ALTERNATIVES

This chapter presents how users in Sinacaban currently cope with the lack of a centralized agricultural system, evaluates existing digital platforms related to farming and trade, and discusses how the proposed project stands out in addressing these limitations.

Several digital initiatives and information systems have been developed to improve market linkages and agricultural coordination among farmers in the Philippines and other developing countries. One of the most notable is the Department of Agriculture's e-Kadiwa program, which aims to connect local producers directly with consumers through an online marketplace. Similarly, the DA eMarketplace provides a national digital platform for trading agricultural products. These systems demonstrate the government's commitment to digital transformation in the agricultural sector. However, studies have shown that their utilization in rural communities remains low due to limited internet connectivity, lack of localized features, and low digital literacy among farmers (Briones et al., 2023; Pastolero & Sassi, 2022).

In practice, many smallholder farmers rely on social media platforms such as Facebook Marketplace and community group chats to sell their products. These tools provide accessibility and a wide reach but lack structured features for managing seasonal harvests, recording data, or tracking buyer reliability (Briones et al., 2023; Pastolero & Sassi, 2022). Transactions are often informal and unmonitored, which makes it difficult for farmers and agricultural officers to track prices, volumes, and the overall flow of goods. Similar observations were reported by (Ma, Renwick, & Singh, 2024), who found that unstructured digital exchanges tend to exclude institutional actors and limit long-term coordination. This situation reinforces the need for dedicated agricultural information systems that combine accessibility with specialized functionalities to support coordination, transparency, and efficient market linkages (Abdul Latif Jameel Poverty Action Lab (J-PAL), 2023; Qiang, Kuek, Dymond, & Esselaar, 2012).

Beyond the Philippines, other digital platforms for agriculture such as FarmLogs, AgriDigital, and AgroTech have been developed in other regions to address similar challenges. These tools focus on record-keeping, logistics management, and price transparency. While they have achieved success in improving efficiency and decision-making, they are often designed for commercial-scale farms or regions with stable internet access. As a result, they are less adaptable to the technological and infrastructural conditions of rural barangays in the Philippines, where most farmers operate on small scales and depend on intermittent connectivity (Ma et al., 2024).

Previous research emphasizes that information asymmetry and weak digital connectivity remain major barriers to rural farmers' productivity and profitability (Briones et al., 2023). (Mopera, 2016) reported that post-harvest losses in the country can reach up to 42% for vegetables and 28% for fruits, largely due to inadequate logistics and poor coordination among market actors. The World Bank (World Bank, 2013) and Qiang et al. (Qiang et al., 2012) also noted that digital technologies can significantly improve agricultural performance when designed with inclusiveness and local accessibility in mind. However, many existing applications focus primarily on market transactions and overlook the seasonal nature of farming, where planning and timing are as critical as selling.

The proposed *Sinacaban Digital Farmers' Seasonal Produce System* aims to address these gaps by integrating the strengths of previous systems while responding to the unique challenges faced by local farmers in Barangay Sinacaban. Unlike existing national or private platforms, this system is designed for barangay level use and emphasizing simplicity. It also introduces a three way coordination model that connects farmers, buyers, and the Department of Agriculture through a unified, cloud-based platform. By combining market information, crop scheduling, and institutional monitoring, the system seeks to enhance productivity, reduce post-harvest losses, and strengthen digital connectivity in the local farming community.

Sinacaban's smallholder farmers currently rely on platforms like e-Kadiwa and DA eMarketplace, both of which are designed to help with sales. However, these platforms don't really face some of the most pressing issues. They don't provide barangay-level crop scheduling, for instance, or track harvest volumes in detail, and they certainly don't offer a way to monitor buyer commitments. Informal ways, like Facebook Marketplace, are also problematic. They don't generate the structured data that the local Department

System (License)	Platform	Brief Description	Basic Functionalities	Relation to the Proposed System
DA e-Kadiwa (Government platform)	Web	National online marketplace connecting accredited producers and consumers.	Online product catalog, ordering, basic logistics coordination for partner institutions.	Useful for national-level transactions but not tailored to barangay-level crop planning or local coordination.
DA eMarketplace (Government platform)	Web	Digital platform of the Department of Agriculture for trading agricultural goods.	Posting of products, price display, limited analytics and reporting.	Does not provide synchronized planting schedules or barangay-level stock monitoring.
Facebook Marketplace / Group Chats (Proprietary; Meta Platforms)	Web & Mobile App	General-purpose marketplace and messaging channel used informally by farmers.	Posting of items, chat-based negotiation, basic search by location.	Accessible and familiar, but unstructured; lacks data recording, seasonal crop tracking, and institutional view.
Commercial farm-management apps (e.g., FarmLogs, AgriDigital; Commercial / Subscription)	Web & Mobile App	Tools designed for larger farms in developed contexts.	Field records, logistics management, inventory tracking, sometimes payment and contract management.	Feature-rich but require stable connectivity and subscription fees, and are not localized for smallholder farms.

Table 1. Existing digital platforms related to the proposed system

of Agriculture needs for planning purposes, and they don't allow for stock aggregation. Commercial solutions, on the other hand, are often overly complicated, costly, and assume that users have both a reliable internet connection and advanced digital skills. A notable deficiency is the absence of a localized, integrated platform capable of facilitating synchronized seasonal crop planning at the barangay level, offering real-time produce visibility for buyers, and enabling institutional monitoring by the agricultural office. Although existing tools address certain aspects individually, none present a unified, user-friendly, cloud-based solution tailored to the needs of rural municipalities.

PROJECT DETAILS

Overview

The Digital Farmers' Produce System is a cloud-based web application that supports market linkages in Municipality of Sinacaban. It connects four main groups of users:

- **Farmers** manage their produce listings, indicate harvest windows and quantities, and submit requests for resources such as seeds or fertilizers.
- **Buyers** search and filter available products, view details, and contact farmers for possible transactions.
- **Agriculture Officers** monitor barangay-level stock information, respond to resource requests, publish advisories, and generate simple summaries for decision-making.
- **Administrator** handles user and configuration management.

The system will run in a shared cloud hosting environment and is designed to be usable on both desktop and mobile browsers, considering intermittent connectivity and varying levels of digital literacy among users.

Project Framework

The project follows the Rapid Application Development (RAD) framework. RAD focuses on quick prototyping, iterative design, and active user feedback throughout the development process. This approach ensures that the system is built around the real needs of farmers and agricultural officers while allowing flexibility for adjustments during development.

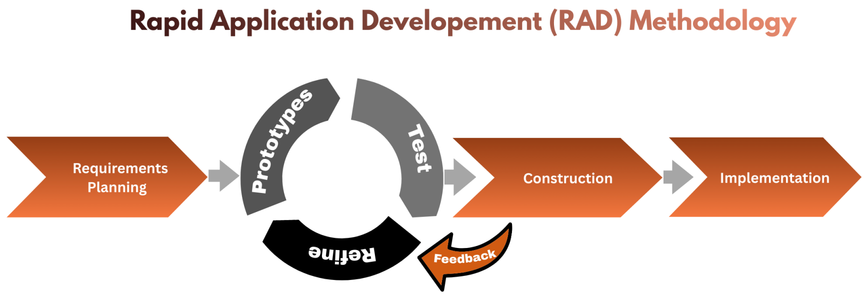

Figure 1. Rapid Application Development (RAD)

Phases of the RAD Framework

1. **Requirements Planning** – Gather data from farmers, buyers, and agricultural officers through interviews and surveys to identify key functions and limitations.
2. **User Design** – Create mockups, user flows, and wireframes that visualize the main system processes such as posting, searching, and monitoring produce.
3. **Construction** – Develop and integrate the web system using cloud-based technologies. Continuous testing and user feedback will refine the system's usability.
4. **Cutover** – Deploy the system for pilot use in Barangay Sinacaban, provide training to users, and evaluate its performance and adoption rate.

System Models

Infrastructure Architecture

Figure 2 shows how the Digital Farmers' Produce System is deployed and accessed by its users. Farmers, buyers, and Department of Agriculture (DA) officers use standard web or mobile browsers to connect to the system through a secure HTTPS connection. Their requests pass through a centralized web and application server, which manages

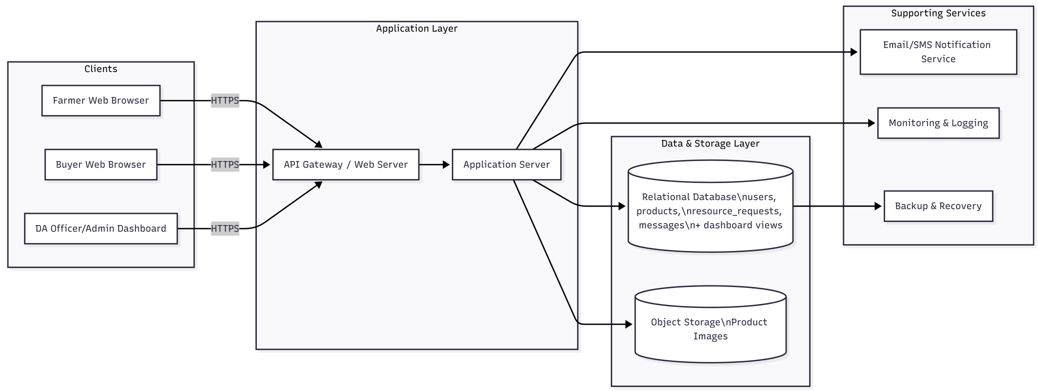

Figure 2. Infrastructure Architecture

user sessions, input validation, and core business logic.

In addition, the system is implemented as a Progressive Web App (PWA). Service workers operate in the background to cache key application resources and frequently accessed data, allowing the system to function even during unstable or unavailable internet connections. When connectivity is restored, the service workers automatically synchronize locally stored data with the cloud-based MySQL database and image storage. A backup and monitoring component continuously copies database records and media files to secure backup storage and monitors system health. Overall, the architecture highlights a cloud-hosted, centrally managed system that remains accessible, reliable, and resilient even in low-connectivity environments.

Entity Relationship Diagram

Figure 3 presents the logical data model of the system. The central *users* table stores all account information and is linked to several core entities. Each farmer can own many products, and each product belongs to a *product_categories* record for standard classification. *messages* connect one user as the sender and another as the receiver, capturing private communication between farmers, buyers, and DA officers. The *resource_requests* entity records farmers' requests for assistance, with DA officers linked as the processors of those requests. Summary or reporting views such as *v_available_by_category* and *v_resource_requests_metrics* provide aggregated information

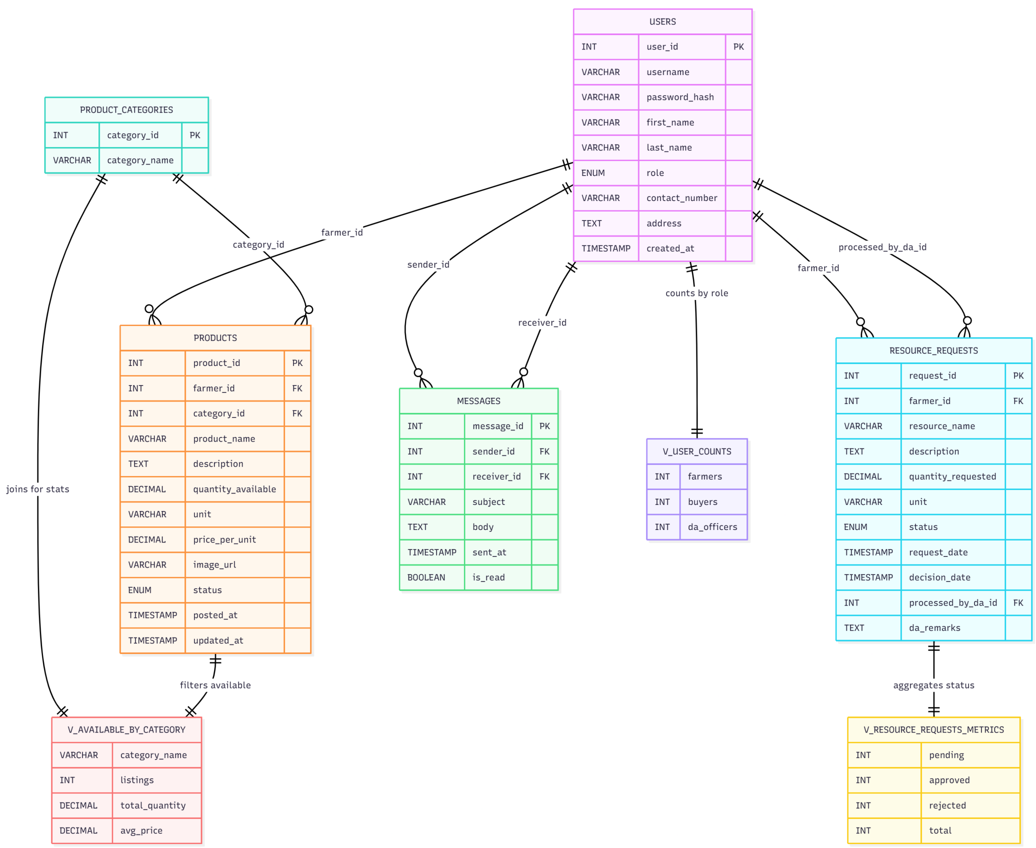

Figure 3. Entity Relationship Diagram (Logical View)

for dashboards, like total available quantity per category or counts of approved and pending requests. This ERD shows how the database supports inventory, communication, and monitoring in a structured way.

Use Case Diagram

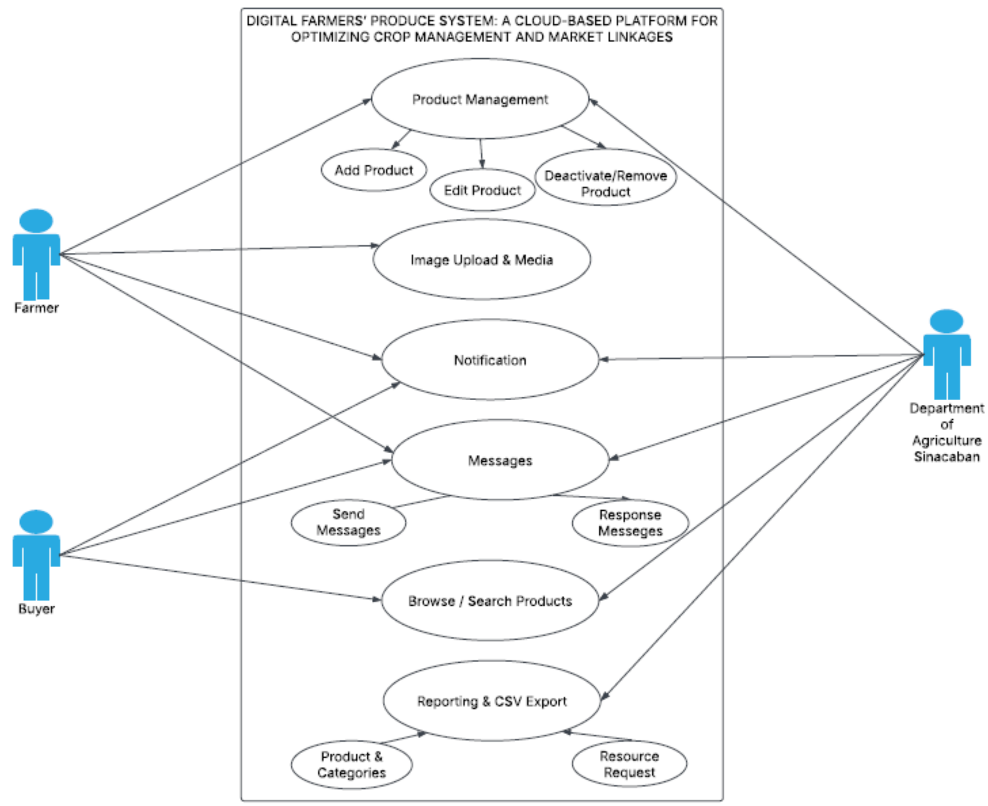

Figure 4. Use Case Diagram

Figure 4 summarizes what each role can do in the system. The farmer actor interacts with use cases related to product management (add, edit, and deactivate products), upload of product images and media, sending messages, receiving notifications, and submitting resource requests. The buyer actor can browse and search products, send messages to farmers, and receive responses and notifications. The Department of Agriculture Sinacaban actor focuses on oversight and support: they manage resource requests, respond to messages, send announcements or notifications, and generate reports or CSV exports. Together, these use cases show that the system covers the full cycle from posting produce and discovering products to coordination with the DA and generation of monitoring reports.

Context Diagram (Level 0)

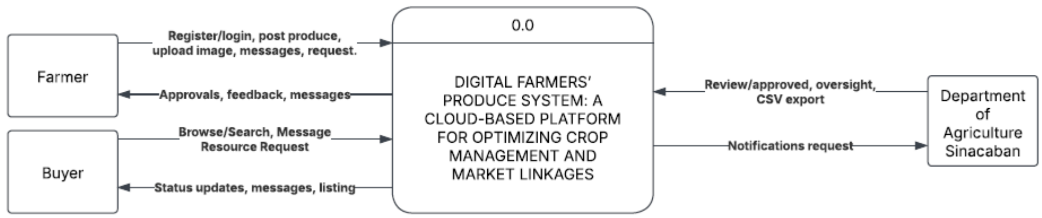

Figure 5. Context Diagram (Level 0)

Figure 5 gives a high-level context view of the Digital Farmers' Produce System as a single process interacting with external entities. Farmers send data such as registration details, product posts, images, messages, and resource requests into the system. Buyers provide registration information, browsing and search requests, and messages. The Department of Agriculture Sinacaban sends review decisions, oversight inputs, and notification content, and in return receives CSV exports, status summaries, and feedback from the system. This diagram clearly defines the system boundary: everything inside the central DFPS block is part of the software, while farmers, buyers, and the DA office are external actors exchanging information with it.

Data Flow Diagram (Level 1)

Figure 6 decomposes the system into major internal processes and data stores. The 1.0 Auth & Session process validates user credentials and manages sessions using the *User* data store. 2.0 Product Management handles adding, editing, and deactivating products, working with the *Products & Categories* data store. 3.0 Requests Workflow and *D3 Resource Requests* manage the life cycle of farmers' resource requests, from submission to approval or rejection by DA officers. 4.0 Notification and *D4 Notifications* are responsible for recording and delivering announcements and alerts to users. 5.0 Image

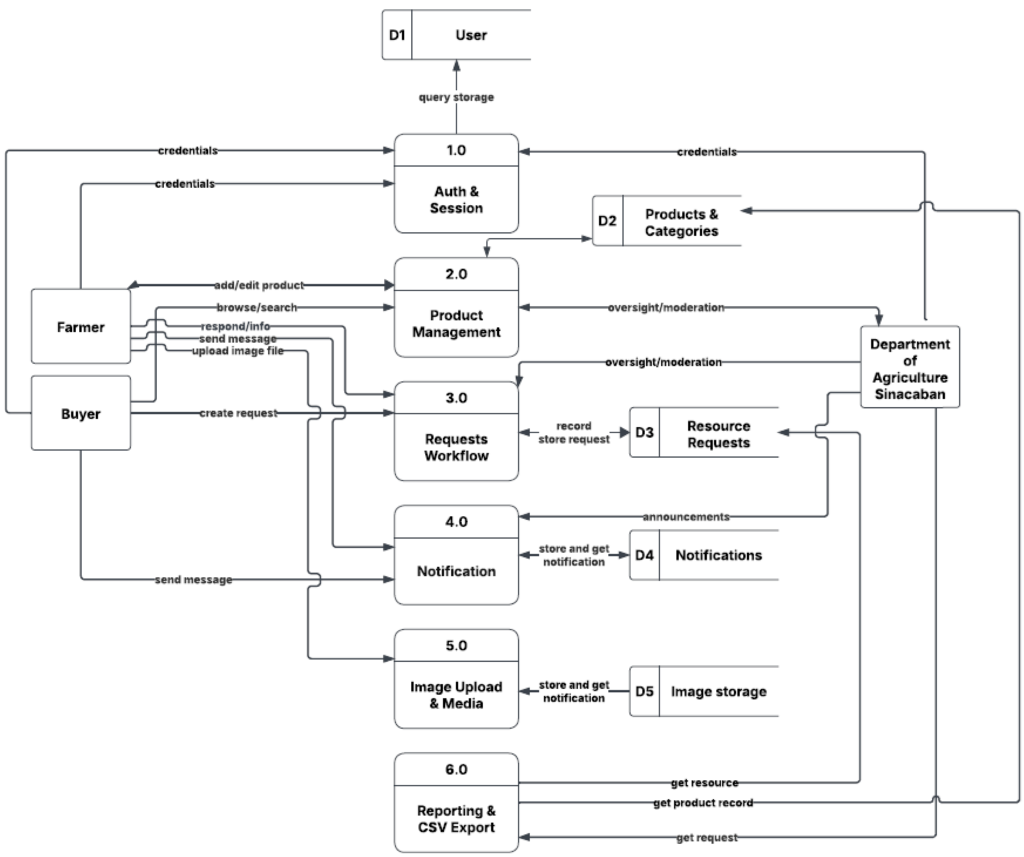

Figure 6. Data Flow Diagram (Level 1)

Upload & Media interacts with *D5 Image Storage* to store and retrieve product images. Finally, 6.0 Reporting & CSV Export reads from the relevant data stores to generate reports and downloadable CSV files. The flows between farmers, buyers, DA officers, and these processes show how information moves through the system from login up to reporting.

Wireframe

Login Page

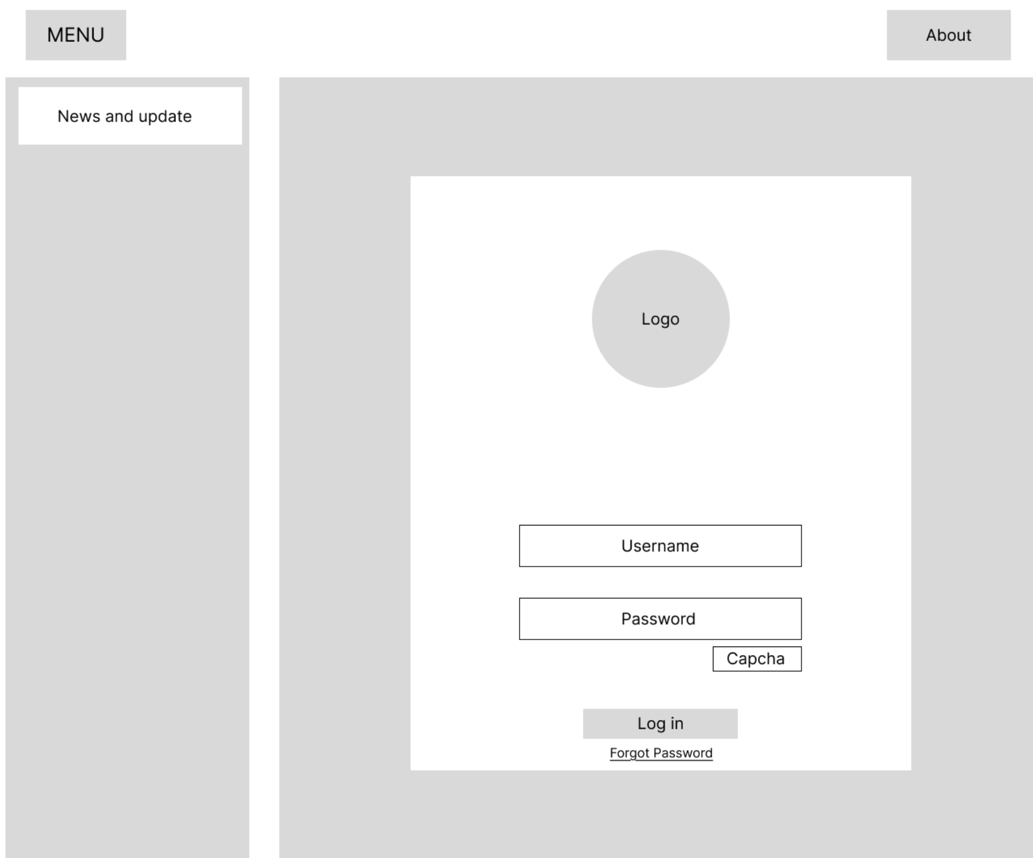

Figure 7. Log In Form

Figure 7 shows the login page wireframe. The system logo is centered, and the form contains fields for Username, Password, and a Captcha to prevent bots. The *Log in* button submits the credentials, while *Forgot Password* allows account recovery. An *About* button at the top-right gives a short description of the system for first-time users.

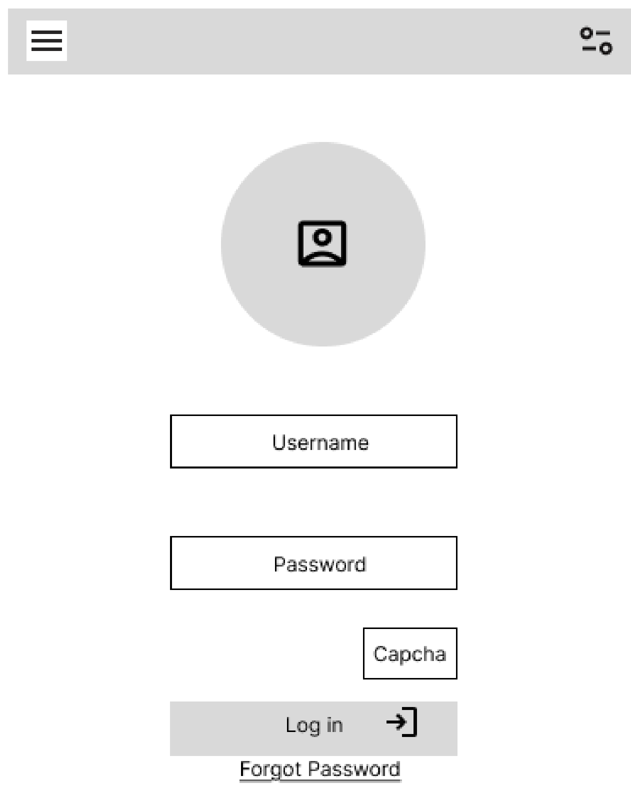

Figure 8. Log in Form Mobile

Registration Page

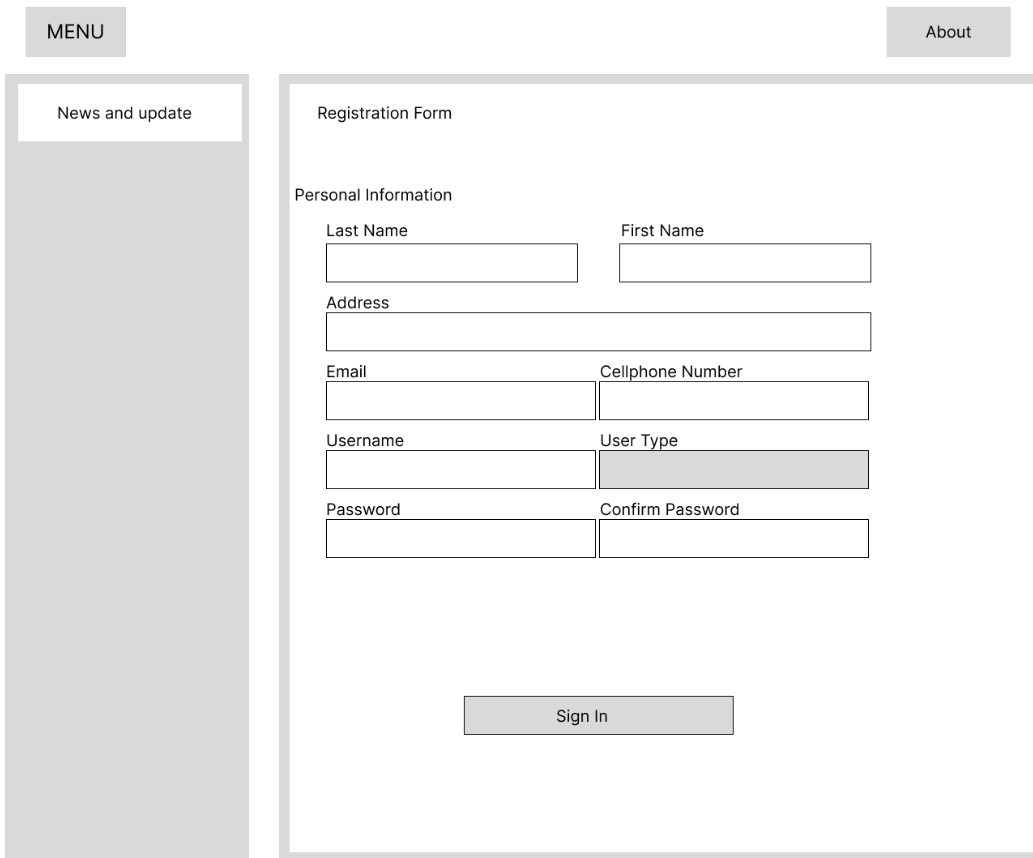

Figure 9. Registration Form

Figure 9 shows the registration page used by new users (farmers or buyers) to create an account. It collects personal information such as Last Name, First Name, Address, Email, Cellphone Number, Username, Password, and User Type (Farmer, Buyer, DA staff). The *Sign In* button submits the form and creates the account once the data is valid.

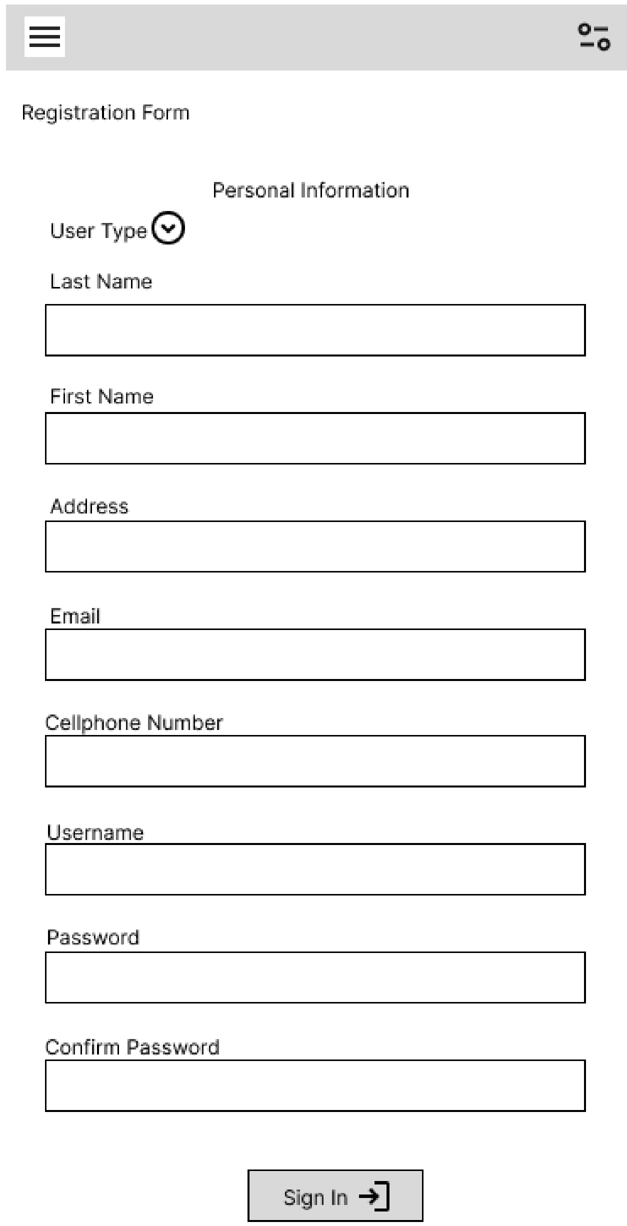

Figure 10. Registration Form Mobile

Farmer Home Page

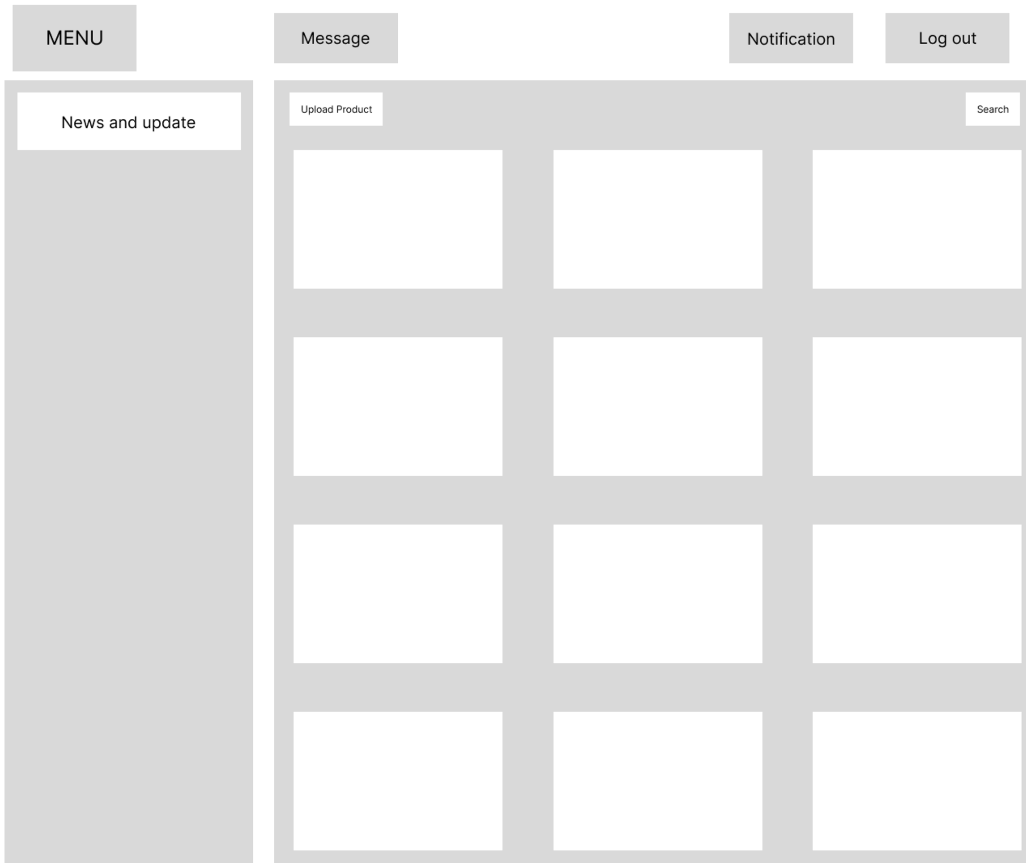

Figure 11. Farmers Home Page

Figure 11 shows the farmer home page. The main area displays a grid of product cards representing the farmer's uploaded products. An *Upload Product* button at the top-left lets farmers add new listings (photo, price, quantity, and other details). A Search box at the top-right helps farmers quickly locate specific products. This page focuses on managing the farmer's own produce.

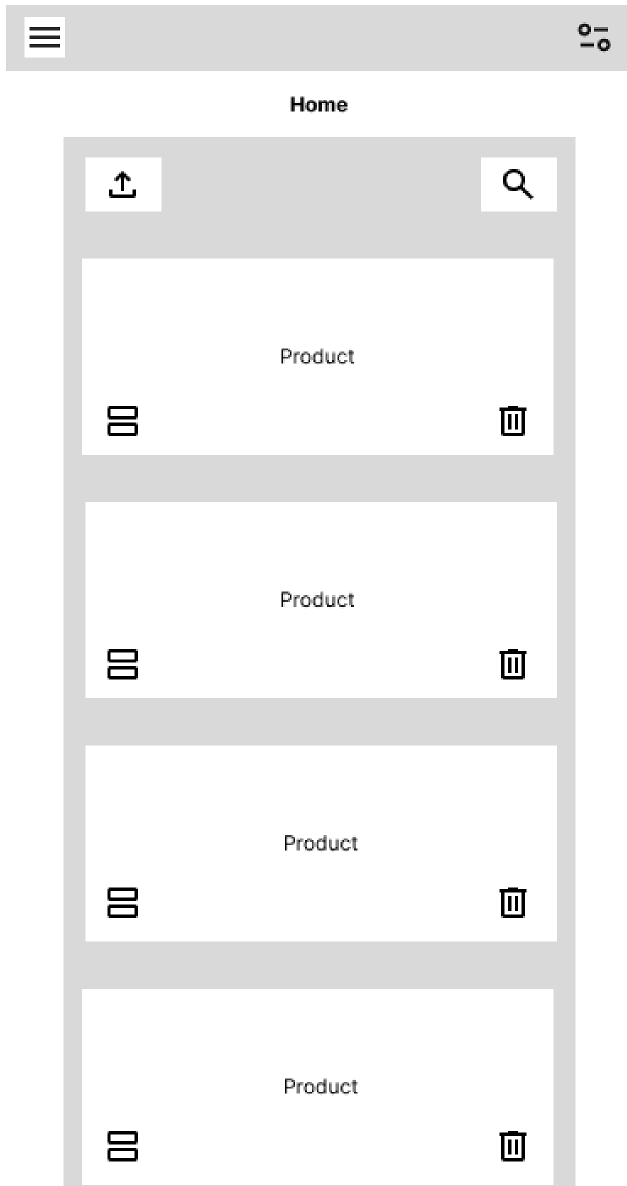

Figure 12. Farmers Home Page Mobile

Buyer Home Page

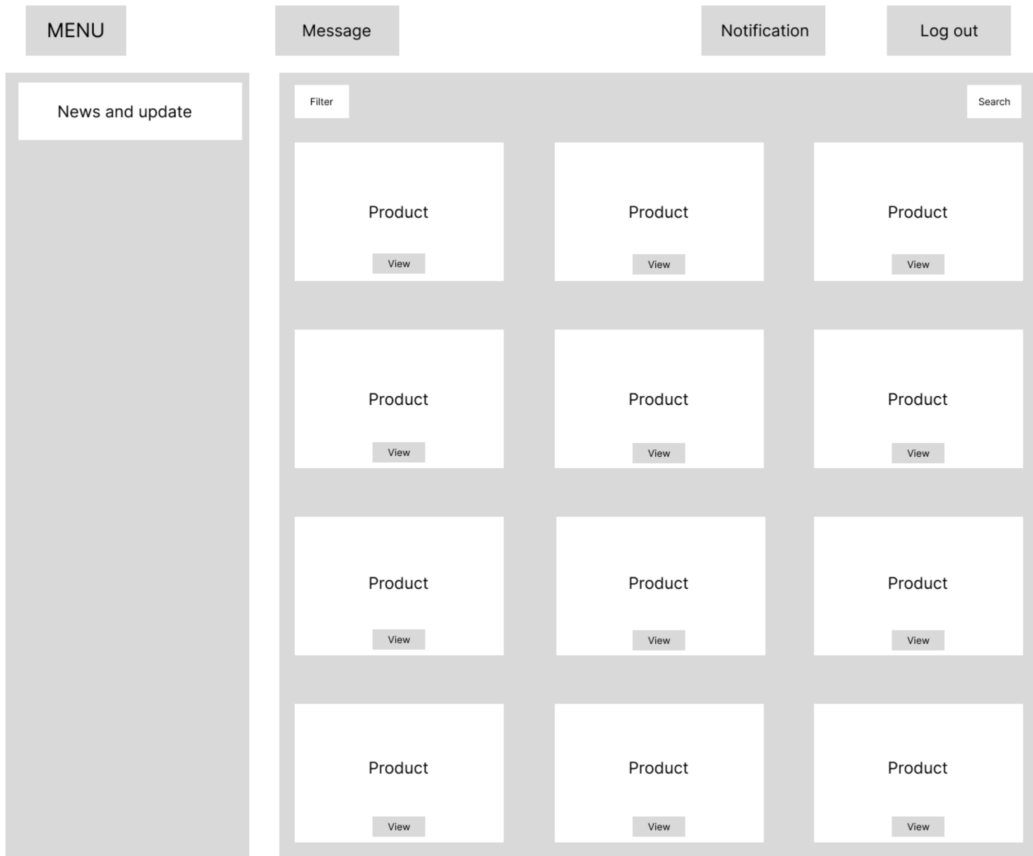

Figure 13. Buyer Home Page

Figure 13 shows the buyer home page, which uses a similar grid layout but from the buyer's point of view. Each card shows a product with a *View* button for detailed information (price, quantity, farmer, contact). A *Filter* button lets buyers narrow products by criteria such as crop type or availability, while the Search box supports keyword search. This page is designed for exploring and selecting products to buy.

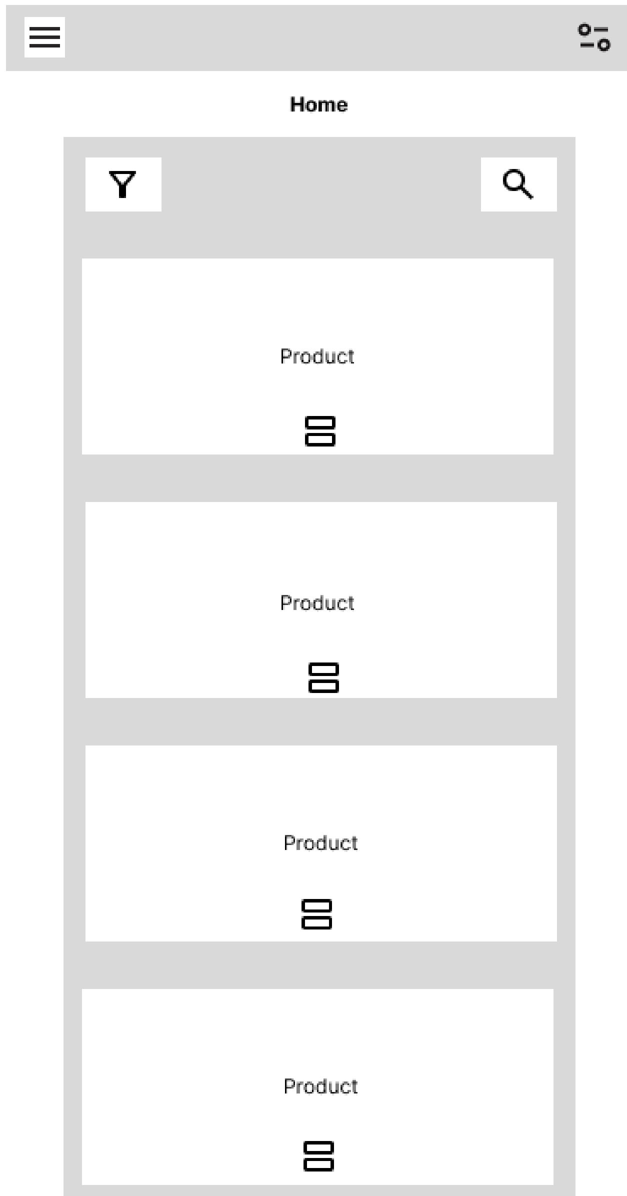

Figure 14. Buyer Home Page Mobile

Department of Agriculture Home Page

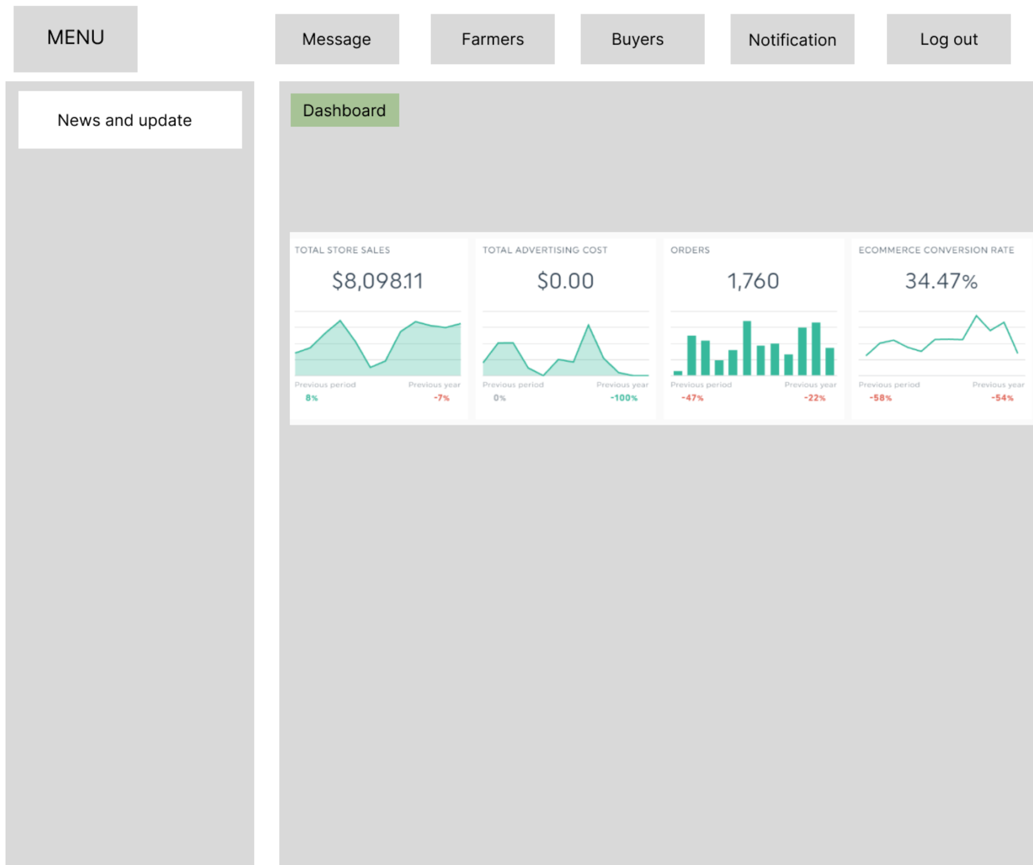

Figure 15. Department of Agriculture Home Page

Figure 15 shows the DA home page dashboard. The main area presents summary tiles with key performance indicators such as total store sales, advertising cost, number of orders, and e-commerce conversion rate. Each tile includes a small chart comparing the current period with the previous period or year, helping DA staff quickly monitor platform performance and agricultural activity.

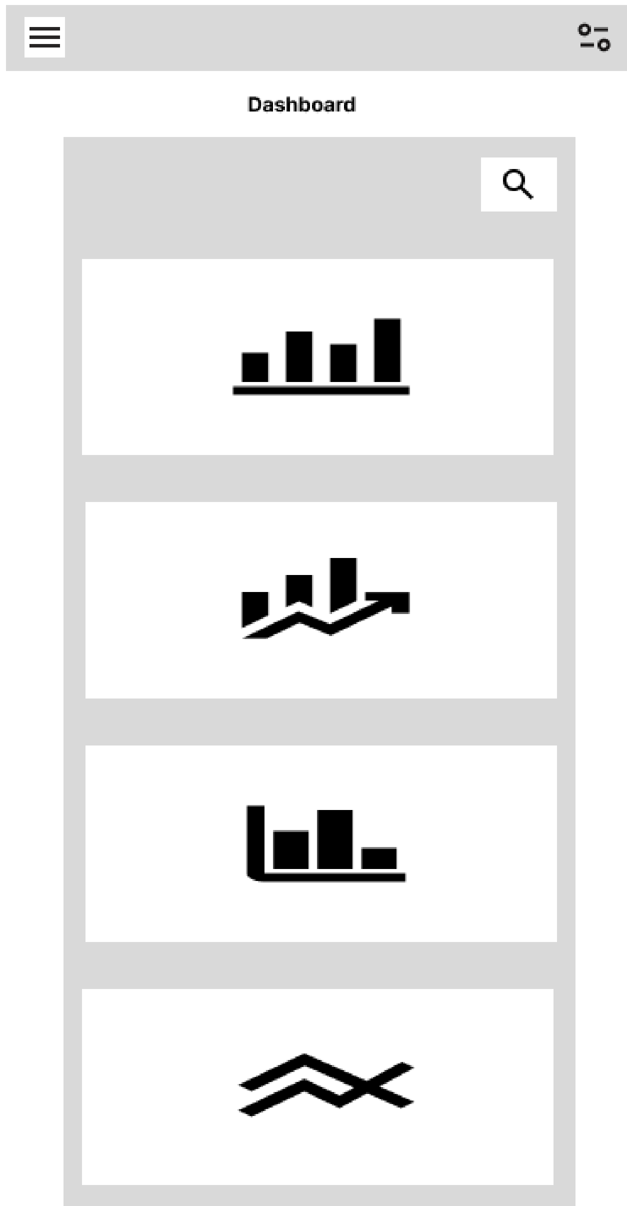

Figure 16. Department of Agriculture Home Page

DA Farmers Page

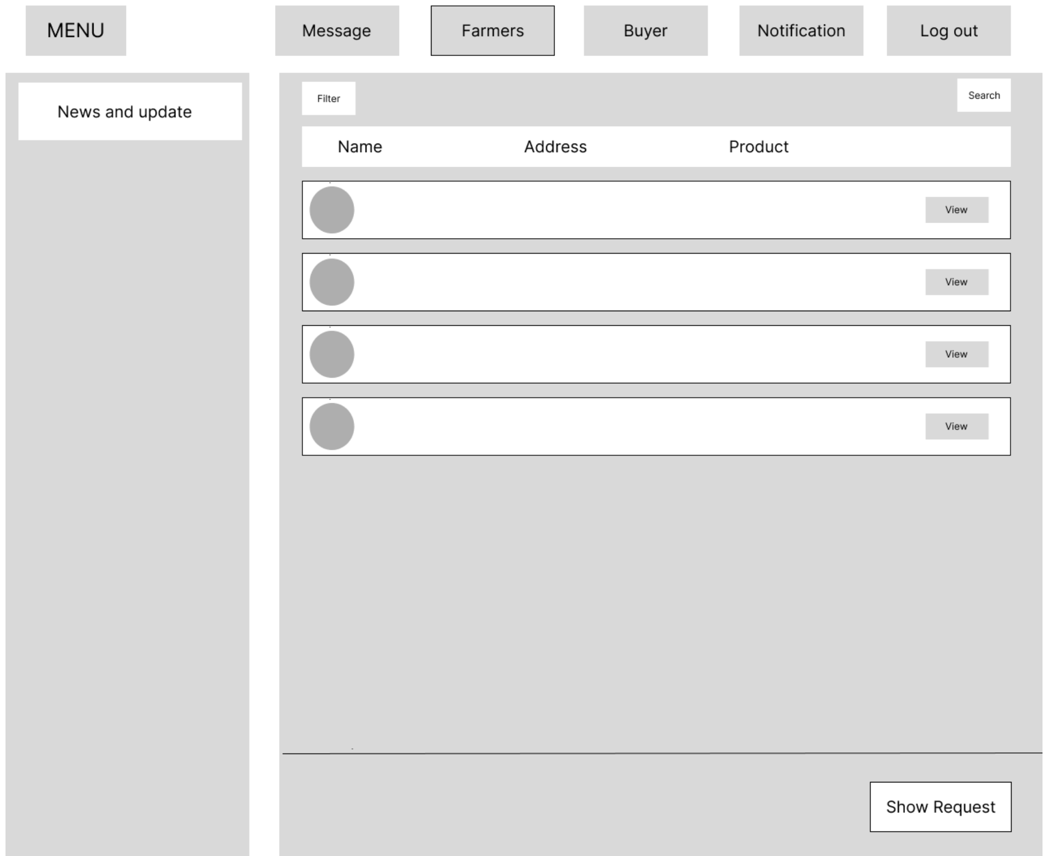

Figure 17. DA Farmers Page

Figure 17 shows the DA farmers page, which lists registered farmers. Columns include Name, Address, and a summary of Products sold. Each row has a *View* button to open the farmer's profile and details. A *Filter* button and Search box allow DA staff to find farmers based on criteria such as location. At the bottom, a *Show Request* button displays pending requests from farmers (e.g., registration approval or assistance).

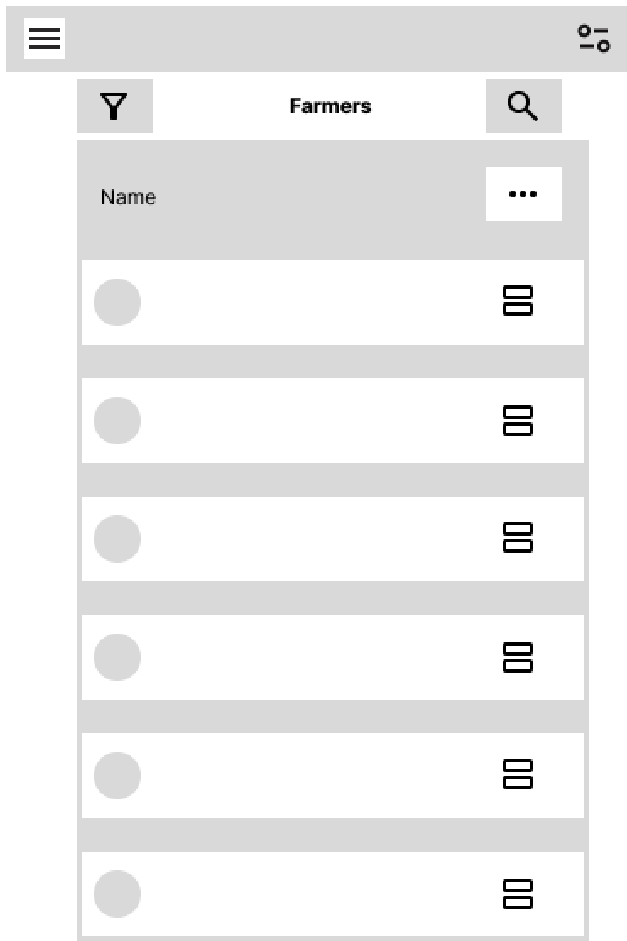

Figure 18. DA Farmers Page Mobile

DA Buyers Page

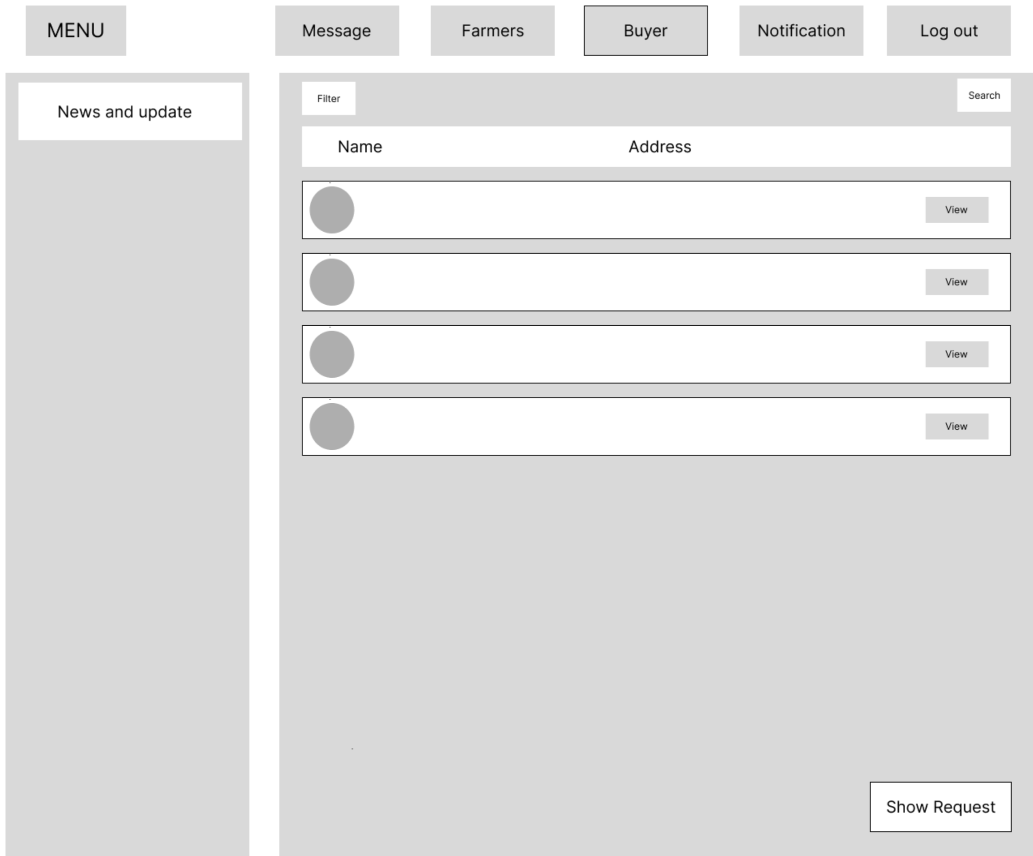

Figure 19. DA Buyers Page

Figure 19 shows the DA buyers page, which has a structure similar to the farmers page but for buyers. It lists Name and Address for each buyer, with a *View* button to inspect the buyer's profile. Filter, Search, and *Show Request* tools support the management of buyer requests and registrations.

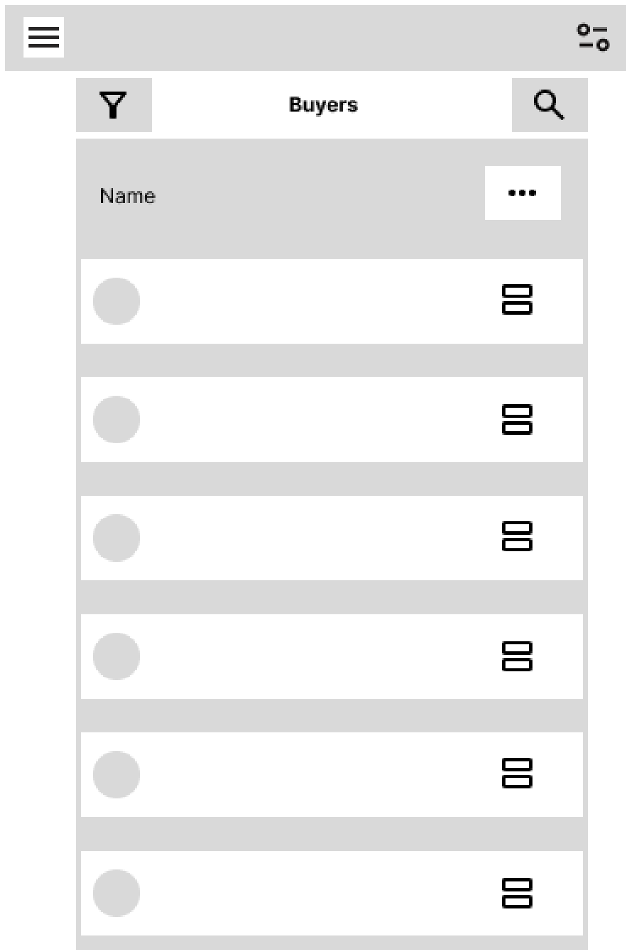

Figure 20. DA Buyers Page Mobile

Message Page

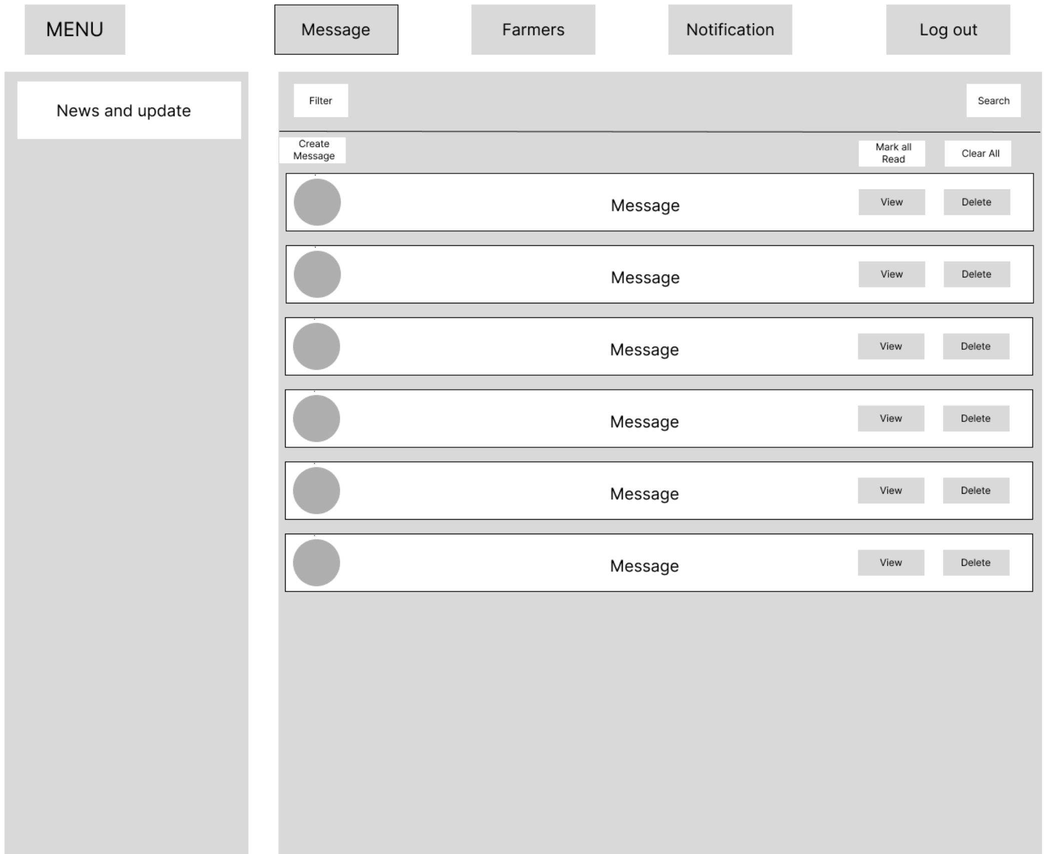

Figure 21. Message Page

Figure 21 shows the messaging page used for communication between users (for example, farmer–buyer or farmer–DA). At the top are tools for *Filter*, *Search*, *Mark all Read*, and *Clear All*. A *Create Message* button lets the user start a new conversation. The main list displays individual messages with the sender's profile image, a short text preview, a *View* button to open the full message, and a *Delete* button to remove it from the inbox.

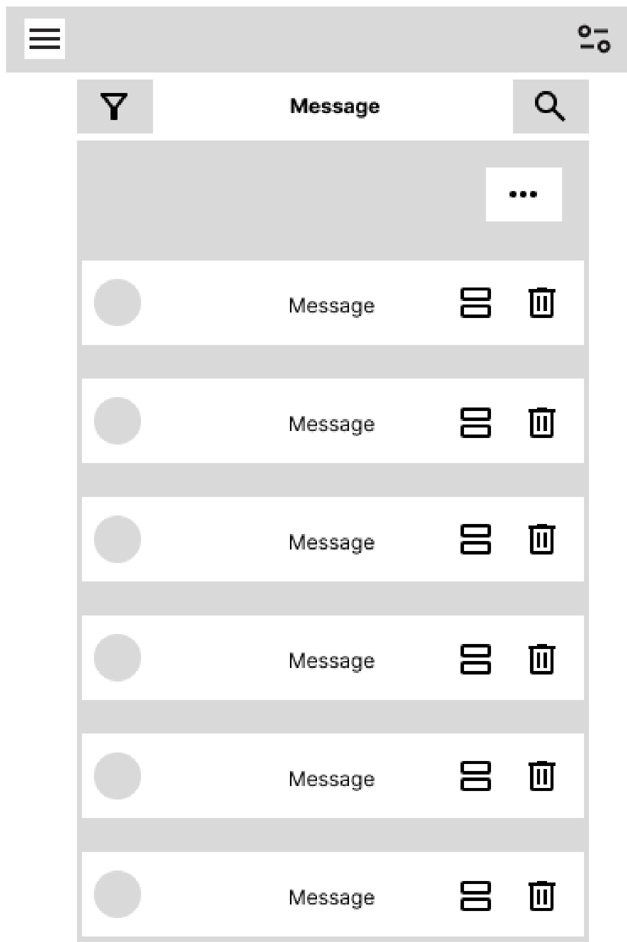

Figure 22. Message Page Mobile

Notification Page

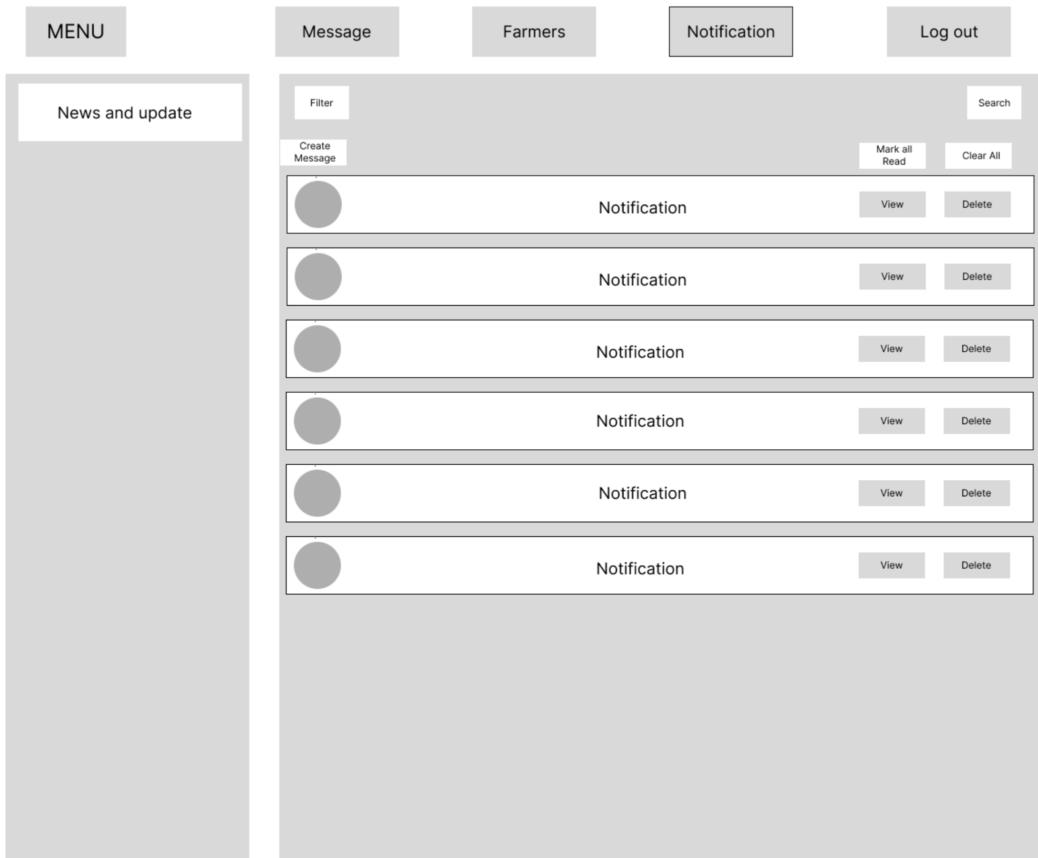

Figure 23. Notification Page

Figure 23 shows the notifications page, which has a similar layout to the message page but is dedicated to system notifications (for example, “Your request was approved” or “New order received”). The tools include *Filter*, *Search*, *Mark all Read*, and *Clear All*. Each notification can be viewed in detail or deleted, helping users stay updated on important system events without mixing them with personal messages.

Materials/Technologies Used

The system was built on a PHP MySQL stack suitable for shared hosting and local government or community level deployment.

Backend Runtime and Framework

- PHP 8.x is used as the primary backend runtime.
- The Flight microframework handles routing and controller dispatch. Routes are defined with `Flight::route`, mapping HTTP paths to controller functions responsible for processing requests and rendering responses.
- Composer manages dependencies and PSR-4 autoloading, helping keep the codebase organized and maintainable.

Frontend Technologies

- The user interface is built with HTML5, CSS, JavaScript, and Bootstrap 5.
- Bootstrap's grid system and components are used to create responsive forms, tables, and cards that adapt to various screen sizes, from smartphones to desktop monitors.
- Basic accessibility practices are applied, including descriptive labels, logical tab order, and sufficient color contrast.

Database

- A MySQL database stores structured data for users, roles, product categories, product listings, messages, resource requests, and audit logs.
- Tables are indexed on commonly searched fields such as product name, category, and barangay to support efficient filtering and search.
- Foreign keys and constraints are used where appropriate to maintain referential integrity.

Cloud Infrastructure

- The application is deployed on GoDaddy shared/cloud hosting with an associated domain.
- HTTPS/SSL is enabled, so all traffic between clients and the server is encrypted.
- Hosting features such as automated backups and basic resource monitoring are used to support availability and recovery targets.

Development Tools

- A PHP-compatible code editor or IDE is used for development.
- Git manages source code versions and supports tracking of changes and rollbacks when necessary.
- Simple task lists or lightweight project management tools are used to follow progress on features, bugs, and enhancements.

These technologies were chosen because they are widely supported, relatively low-cost, and appropriate for organizations operating under limited infrastructure and budget.

System Design

The system design covers both what the application does (functional design) and how it behaves under various conditions (non-functional design).

Functional Design

The main functional components of the system are:

- **User Management and Roles** – Users can register and log in, with access restricted based on their role (farmer, buyer, agriculture officer, or admin). Each role sees role-specific menus and dashboards.

- **Product Categories and Listings** – Farmers can create, edit, view, and archive product listings. Each listing contains fields such as crop name, category, quantity, unit, price, image, barangay, status, and an optional harvest window. Categories are managed centrally to support consistent filtering.

- **Buyer Search and Discovery** – Buyers can browse and search for products using filters such as category, crop name, location, price range, and harvest date. Results can be paginated and sorted to make it easier to find relevant listings.

- **Messaging and Inquiries** – A simple messaging module allows buyers and farmers to send and receive messages related to listings. Users have an inbox-style interface, with read/unread indicators for each message or conversation.

- **Resource Requests** – Farmers can submit requests for resources (for example, seeds, fertilizers, or technical support). Agriculture officers can review these requests, approve or reject them, add remarks, and change their status. Farmers can see updates to the status of their requests.

- **Dashboards and Metrics for Officers** – Agriculture officers have dashboards that summarize key data such as total active listings, counts per category, number of farmers with active postings, and statuses of resource requests. These views help them monitor agricultural activity at the barangay level.

- **CSV Export** – Authorized users can export product listings to a CSV file, based on any currently applied filters. This enables further offline analysis in spreadsheet software and simplifies reporting for offices and programs.

- **Validation, Feedback, and Logging** – Server-side validation is applied to important forms, with user-friendly error messages displayed when data is missing or invalid. Flash messages indicate success or failure for actions like creating or updating records. Basic logging is implemented for key actions to support governance and troubleshooting.

- **SMS Notification** – The system supports SMS-based alerts for farmers without smartphones. Text messages are used specifically for system announcements and Department of Agriculture (DA) subsidy notifications, ensuring that farmers receive important official updates even in areas with limited internet connectivity or when using basic mobile phones.

Non-Functional Design

Several non-functional qualities were built into the design:

- **Usability and Accessibility** – Screens are kept clean and simple, with clear labels and minimal required fields. Navigation is consistent across roles, and forms are designed to work well on mobile devices, supporting users with low to moderate digital literacy. The system includes a language toggle with a choice of Bisaya, allowing users to view the interface in their preferred language. To further improve accessibility, the design reduces reliance on text through the use of clear icons, visual cues, and simplified labels, making the system easier to understand for users with varying literacy levels.

- **Performance** – Page templates and queries are optimized to perform acceptably on shared hosting and under intermittent connectivity. Commonly queried fields are indexed, and unnecessary data loading is avoided to keep response times reasonable.

- **Security** – All access to the system is through HTTPS. Passwords are stored using secure hashing algorithms. Role checks are enforced on the server for each protected route. Input validation, prepared statements, and output escaping are used to reduce vulnerabilities such as SQL injection and cross-site scripting.

- **Availability and Backup** – Database backups are scheduled regularly via the hosting provider, helping ensure that data can be restored if needed. The deployment is structured so that recovery can be done within a practical timeframe in case of failure.

- **Maintainability** – The project uses clear folder structures, consistent naming, and small controller and model classes. Routes are centralized, and configuration is separated from code. This approach simplifies future updates, bug fixes, and onboarding of new developers.

Implementation

Figure 24. Gantt Chart

Figure 24 shows that the project begins with Requirements Planning in November (*Initiation* – dark green), which covers the kickoff, planning, stakeholder interviews, document collection, and note-taking on related literature from November 1–27. This is followed by User Design from late November through December (*Planning* – light green), where the researcher synthesizes the gathered information, prepares an outline, writes the full documentation from December 2–10, reviews and revises it from December 11–14, and then conducts the oral defense from December 15–19.

In January and February, Construction (*Execution & Monitoring* – orange) represents the developing the core modules of the system from January 2–21, followed by testing and bug fixing from January 22–February 11, th deployment of the pilot and user training from February 1–14, and user testing and security activities from February 8–21. Finally, the Cutover phase in March (*Closing* – brown) corresponds to User Acceptance Testing based on TAM from February 22–March 6, the review and documentation of results from March 1–14, and preparation for the final defense and submission of all requirements from March 8–28, which formally concludes the project.

Project Assessment

User Testing

Objectives of User Testing

User testing will be conducted to:

1. Verify that each functional requirement of the Digital Farmers' Produce System works as intended.
2. Evaluate non-functional requirements such as usability, performance, reliability, and compatibility.
3. Detect bugs, usability issues, and gaps before deployment.

The overall aim is to ensure that the system is usable and acceptable to its intended users and that no critical defects remain at the time of implementation.

Target Users and Test Environment

User testing will involve representative users from the three main roles:

- **Farmers** – smallholder farmers who will post and manage their produce.
- **Buyers** – local traders, retailers, and institutional buyers.
- **DA / MAO Staff** – personnel from the Municipal Agriculture Office who will monitor listings and requests.
- Planned sample size:
  - 5–8 farmers
  - 4–6 buyers
  - 3–4 DA / MAO staff

Testing will be done using:

- Low-end Android smartphones, to reflect typical devices of farmers and buyers.
- A standard desktop or laptop with a modern web browser, to reflect use by DA staff.

All tests will be conducted on a staging version of the system with test data.

Test Specifications for Functional Requirements

Functional testing will be scenario-based. Each functional requirement will have at least one formal test case with:

- Test ID and name
- Description
- Pre-conditions
- Step-by-step procedure
- Expected result
- Actual result (to be filled during testing)
- Pass/Fail status

Examples of planned scenarios per role:

Farmers

- Register an account and log in.
- Create a new crop listing (with crop name, quantity, price, barangay, and harvest window).
- Edit an existing listing and then deactivate it.
- View messages and reply to a buyer.

Buyers

- Register an account and log in as buyer.
- Search for products using keywords and filters (e.g., crop, barangay).
- Open a listing and send a message to the farmer.

DA / MAO Staff

- Log in and access the dashboard.
- View and update the status of resource requests from farmers.
- View registered farmers and buyers.
- Export the list of active listings to CSV.

A functional requirement will be considered passed when:

- The steps can be completed without system error.
- The observed result matches the expected result.
- At least 90% of users assigned to that role can complete the scenario with minimal assistance.

During testing, problems will be perceived when tasks cannot be completed, when unexpected errors occur, or when users repeatedly hesitate or ask for clarification.

Test Plan for Non-functional Requirements and Acceptance Criteria

Non-functional requirements will also be tested with clear indicators and acceptance criteria.

a. Usability

- Number of tasks completed without assistance.

- Number of observed errors or points of confusion.
- User ratings on usability questions.

Acceptance criteria:

- At least 80% of tasks are completed without guidance.
- Average rating of 4.0 or higher (out of 5) on items related to ease of use and clarity.

b. Performance

- Page load times for login, dashboard, and product lists.
- Response time for common actions (posting, searching, messaging).

Acceptance criteria:

- Main pages load within 3–5 seconds under normal test conditions.
- No excessive delays that prevent users from completing tasks.

c. Reliability

- Number of crashes or unhandled system errors during testing.
- Frequency of repeated failures on the same function.

Acceptance criteria:

- No critical crashes while executing the planned test scenarios.
- All error messages presented to users are understandable and non-technical.

d. Compatibility / Responsiveness

- Layout behavior on mobile devices and desktops.
- Visibility of labels, buttons, and input fields at common screen sizes.

Acceptance criteria:

- No overlapping or hidden controls on tested devices.
- Key pages remain usable on low-resolution screens.

If any acceptance criterion is not met, the issue will be logged, prioritized, and scheduled for fixing before deployment.

Bug Detection, Tracking, and Regression Testing

Bugs and usability issues will be detected through:

- Developer testing prior to user testing.
- Observations during user sessions.
- Error messages and logs from the application.

All issues will be recorded in a defect log with:

- Unique defect ID.
- Date reported.
- Module or page affected.
- Severity (critical, major, minor).
- Steps to reproduce.
- Expected vs. actual result.
- Assigned developer.
- Status (open, in progress, fixed, closed).

To minimize remaining bugs, the project will follow this sequence:

1. **Developer Testing** – unit-level checks and manual walkthroughs after each feature is implemented.

2. **System Testing** – execution of all functional and non-functional test cases internally.
3. **User Testing** – execution of real-world scenarios by target users.
4. **Fixing and Verification** – defects are prioritized, fixed, and re-tested.

After each set of fixes, regression testing will be done by:

- Re-running test cases related to the affected modules.
- Re-testing core workflows (login, posting, search, requests, messaging).

Before deployment, the target condition is:

- No remaining critical defects.
- No remaining major defects in core workflows.
- Only low-impact minor issues, if any, left open for future enhancement.

User Acceptance Testing Using the Technology Acceptance Model (TAM) – Planned

Purpose User Acceptance Testing will use the Technology Acceptance Model (TAM) to assess whether farmers, buyers, and DA staff are likely to accept and use the system. The focus will be on four constructs:

- Perceived Usefulness (PU)
- Perceived Ease of Use (PEOU)
- Attitude Toward Using (ATU)
- Behavioral Intention to Use (BI)

The results will indicate whether the system is not only technically correct but also acceptable and meaningful to its intended users.

Survey Instrument A structured questionnaire will be administered immediately after the hands-on user testing. It will use a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Sample items for each construct include:

Perceived Usefulness (PU)

- "Using this system will help me manage my produce and transactions more efficiently."
- "This system will make it easier for me to find buyers or products."

Perceived Ease of Use (PEOU)

- "Learning to use this system will be easy for me."
- "It will be easy for me to become skillful at using this system."

Attitude Toward Using (ATU)

- "Using this system will be a good idea for our community."
- "I will feel positive about using this system in my work."

Behavioral Intention to Use (BI)

- "I intend to use this system regularly once it is available."
- "I will recommend this system to other farmers, buyers, or staff."

The same participants from the user testing phase will be asked to answer the survey so that their ratings are based on direct experience with the system.

Data Analysis Plan After data collection, the following analysis will be done:

- Compute mean scores for each item and each construct (PU, PEOU, ATU, BI).
- Assess internal consistency of each construct (e.g., Cronbach's alpha) to check reliability of the survey items.

- Interpret high mean values (close to 5) as strong acceptance and low values as areas needing improvement.

If one or more constructs show low scores (for example, low PEOU), the project researcher will:

- Review and simplify the interface and workflows.
- Revise labels, instructions, and help text.
- Consider design revisions and, if feasible, a follow-up testing round.

Security Testing

Security Testing Approach and Tools

Security testing will focus on ensuring the confidentiality, integrity, and availability of user data and system functions. The plan includes:

1. **Configuration and Code Review**
   - Confirm that all web traffic uses HTTPS.
   - Check that user passwords are stored using secure hashing mechanisms.
   - Verify the use of prepared statements or an ORM to protect against SQL injection.
   - Inspect the implementation of role-based access control.
2. **Automated Vulnerability Scanning**
   - Use an open-source web security testing tool, such as OWASP ZAP or a similar scanner, on the staging server.
   - Scan for common web vulnerabilities, including:
     - SQL injection
     - Cross-site scripting (XSS)

   - Insecure cookies and sessions
   - Missing or weak security headers

3. **Manual Security Checks**
   - Attempt to modify URL parameters (e.g., record IDs) to access data belonging to other users.
   - Enter specially crafted input (e.g., script tags) to test for reflected or stored XSS.
   - Observe error messages to ensure they do not expose database details or system configuration.

Identifying Security Vulnerabilities

A security issue will be considered present when:

- The automated scanner flags vulnerabilities with high or medium severity.
- A user is able to view or modify data they should not have access to.
- Inputs are accepted and interpreted in ways that indicate injection or XSS risks.
- Error messages reveal internal information such as SQL statements, file paths, or stack traces.

All identified security issues will be added to the defect log with severity, impact, and recommended actions.

Mitigation and Re-testing

For each identified vulnerability, appropriate mitigation steps will be applied, such as:

- Replacing detailed error messages with generic user-friendly messages while logging full details on the server.
- Strengthening password policies (e.g., minimum length and complexity).

- Adding or correcting CSRF tokens in forms that change data.
- Configuring recommended security headers (e.g., Content-Security-Policy, X-Frame-Options, X-Content-Type-Options) where supported.
- Adjusting access control rules to prevent unauthorized access.

After fixes are implemented, both automated scans and manual tests will be repeated as part of security regression testing. The security testing phase will be considered satisfactory when:

- No high-severity issues remain in the scanner reports.
- Manual attempts to bypass access control or inject script fail.
- The system behaves predictably and securely under the tested scenarios.

Results and Discussion

This section presents the key findings and main challenges encountered during the project's analysis, design, and initial development of the Digital Farmers' Produce System.

Summary and Conclusion

Future Work

This project lays the foundation for a localized Digital Farmers' Produce System, but it does not yet exhaust all the possibilities for supporting farmers, buyers, and the Municipal Agriculture Office. Several directions for future work can further improve, expand, and deepen the impact of the system.

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